



Glioblastoma and brain connectivity: the need for a paradigm shift

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Lancet Neurol 2024; 23: 740–48

See [Comment](#) page 655

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Despite substantial advances in cancer treatment, for patients with glioblastoma prognosis remains bleak. The emerging field of cancer neuroscience reveals intricate functional interplays between glioblastoma and the cellular architecture of the brain, encompassing neurons, glia, and vessels. New findings underscore the role of structural and functional connections within hierarchical networks, known as the connectome. These connections contribute to the location, spread, and recurrence of a glioblastoma, and a patient's overall survival, revealing a complex interplay between the tumour and the CNS. This mounting evidence prompts a paradigm shift, challenging the perception of glioblastomas as mere foreign bodies within the brain. Instead, these tumours are intricately woven into the structural and functional fabric of the brain. This radical change in thinking holds profound implications for the understanding and treatment of glioblastomas, which could unveil new prognostic factors and surgical strategies and optimise radiotherapy. Additionally, a connectivity approach suggests that non-invasive brain stimulation could disrupt pathological neuron–glioma interactions within specific networks.

Introduction

Glioblastoma has a bleak prognosis, with a median overall survival of approximately 15 months after surgical diagnosis and a 5-year survival rate below 8%.^{1–3} Efforts to enhance prognosis are ongoing,⁴ but progress to date has led to only marginal improvements in survival.⁵ Although substantial improvements in survival rates have been achieved for other cancers,⁶ the bleak trajectory of glioblastoma prompts a reconsideration of our approach to this disease. Traditionally, research has concentrated on the tumour itself, investigating molecular profiles, genetic mutations, histology,^{7,8} and its microenvironment (including immunological infiltrates and radiomics).^{9,10} This approach mirrors that of other cancers like lung, prostate, breast, and stomach tumours. However, a crucial aspect that is often overlooked is the unique organ host of a glioblastoma: the brain.

The novel field of cancer neuroscience^{11,12} (panel 1) offers a paradigm shift by transitioning from an exclusive focus on the tumour to exploring the intricate interactions between a glioblastoma and the brain. This perspective spans from the synaptic level, at the microscale, to neural networks and their connectivity properties, at the macroscale. Contrary to viewing changes in synaptic activity and brain connectivity as mere consequences of the presence of a glioblastoma, this approach recognises them as pivotal contributors to carcinogenesis, influencing cancer growth and spread. This fundamental change towards a connectivity-based approach has the potential to revolutionise the understanding and management of glioblastoma, enhance prognosis, refine current treatments such as surgery and radiotherapy, and open avenues for novel treatment strategies.

A qualitative analogy is to envision the brain as a railway system, with the structural pathways representing the tracks and the functional interactions representing the amount of traffic on those tracks. This analogy emphasises communication through nodes and edges; however, studies published within the past 5 years have shown that

activity propagates through dynamic waves across the connectome (the comprehensive map of white matter tracts and cortical functional connections).¹⁷ Can understanding the structure and function of the brain shed new light on the pathophysiology of glioblastoma? A positive answer emerges from studies of neurodegenerative diseases and stroke. In several disorders characterised by the accumulation of misfolded proteins, pathology spreads through anatomical pathways and preferentially begins in specific functional networks, indicating a widespread involvement of the connectome.^{18–20} Similarly, even small stroke lesions can disrupt structural and functional brain connections widely, and these disruptions strongly correlate with neurological impairment and recovery of function.^{21,22} Overall, these findings underscore the broader relevance of the connectome paradigm in understanding the pathophysiology across neurological disorders, possibly including glioblastoma.

Our Personal View aims to highlight key aspects of the relationship between brain connectivity and glioblastoma. We review notions on structural and functional connectivity, emphasising the role of this approach in comprehending pathophysiology. We also examine the clinical implications of this paradigm shift, considering its potential impact on personalised prognosis and the development of innovative treatments. The examination of brain connectivity will centre on the in-vivo characterisation of macroscale structural and functional pathways using MRI techniques, specifically diffusion MRI and functional MRI (fMRI).

Structural connectivity and cancer spreading

Structural connectivity, measured by use of diffusion tensor imaging (tractography), can be assessed before neurosurgery to identify eloquent white matter tracts passing through or around the tumour.²³ This information is uploaded onto intraoperative surgical navigation systems to help the neurosurgeon preserve specific

functions linked with those tracts (eg, motor functions linked with corticospinal tract integrity, or linguistic abilities associated with the preservation of the arcuate fasciculus). Thus, it is not surprising that structural connectome alterations have the potential to forecast long-term cognitive function in people with glioblastoma.²⁴

Glioblastoma spreads relentlessly and invariably recurs after surgery, even after total resection. A prediction tool to identify where a recurrence is most likely to occur would provide a crucial advantage in a surgery plan. In 1940, the pioneer in glioblastoma pathology Hans Scherer asserted that identifying the growth pattern of glioblastoma was a paramount research goal in this field.²⁵ This recommendation has remained largely overlooked until recently, when new evidence supported and expanded his observations. Scherer identified, among the mechanisms of spread, the climbing of tumour cells along white matter tracts. This pathophysiological concept has been clarified through various mechanisms of tumour cell migration and invasion, including the involvement of axonal guidance proteins (netrins and semaphorins) and integrins.^{26–28} Therefore, notwithstanding the risk of oversimplifying a complex mechanism of molecular interactions (including recently discovered tumour microtubes),^{29,30} and considering the coexistence of different routes of cancer infiltration, the structural connectome (ie, the comprehensive map of white matter tracts) represents a plausible pathway for the spread of glioblastoma. The logical consequence is that the study of structural connectivity in people with glioblastoma offers a novel opportunity for understanding the route of cancer spreading.

Building on this hypothesis, by use of a whole-brain diffusion-weighted imaging (DWI) approach, Wei and colleagues³¹ have characterised the disruption of the structural connectome in a study of more than 100 patients with glioblastoma. They found that glioblastoma-related connectivity disruptions extend beyond the focal lesion into the normal-appearing brain, even encompassing the contralateral hemisphere. The most likely disrupted tracts were association fibre pathways connecting distant cortical brain regions, hence providing a path for the long-range migration of glioblastoma cells. The disruption of the connectome in brain regions further from the glioblastoma location correlated with overall survival.³¹ When this information was added into models to predict overall survival, the accuracy of predictions significantly improved.³¹ Finally, the distance of connectome disruption was associated with the distance of recurrence (whereby disruption in distant cortical regions meant recurrence could also be far away from the original tumour but might not necessarily recur in the same distant region), emphasising that the migration of tumour cells influences the structural connectome even when tumour signals are undetectable through conventional imaging methods.

We have investigated whether local properties of the connectome at the site of a glioblastoma location also

Panel 1: Glossary of terms

Cancer neuroscience

A new field studying the mutual interactions between cancer and the CNS.^{11,12} These interactions occur at synaptic level and within local networks.

Blood oxygen level-dependent (BOLD) signal

A cornerstone in functional MRI, this technique serves as a non-invasive proxy of neural activity fluctuations.¹³

Functional connectivity

The synchronisation of spontaneous BOLD signal fluctuations between different brain regions during resting state functional MRI.¹⁴ Functional connectivity is used to characterise the hierarchical architecture of large-scale neural networks.

Task activity

Activity-dependent BOLD signal changes during the presentation of specific stimuli or tasks, which unveils the dynamic interplay between specific brain regions engaged in a particular cognitive task.

Diffusion-weighted imaging

A technique based on the Brownian motion of water molecules influenced by tissue microenvironments. By applying gradient pulses in various directions, diffusion-weighted imaging induces a phase shift in the MRI signal proportional to water diffusion rates, enabling the assessment of white matter microstructure. This technique provides information about the integrity and direction of axonal pathways in the brain.

Structural connectivity

Based on techniques such as tractography and provides information about anatomical connections between brain regions, to help elucidate the pathways supporting functional interactions.¹⁵

Tract density index

A local connectivity tool that quantifies the density of white matter fibres encompassing a specific brain lesion (eg, a glioblastoma). Tract density index uses information from a normative structural connectivity template.¹⁶

influence overall survival.¹⁶ We embedded the core and oedema of the lesions into a normative connectome atlas of a healthy brain, and developed the tract density index, which measures the number of fibres encompassing the lesion at each location in the brain. We hypothesised that, if the connectome is the biological substrate for cancer spreading, then glioblastoma lesions located within regions that contain a higher density of white matter fibres would be associated with lower survival. Accordingly, patients with lesions at brain locations with high tract density index had a lower survival rate than patients with low tract density index (Kaplan–Meier log rank $p=0.03$). The tract density index was computed non-invasively on

	Tumour	Number of patients (mean age [SD])	Number of controls (mean age [SD])	Framework approach	Measure	Main findings	Overall survival
Salvalaggio et al (2023) ¹⁶	Glioblastoma	Discovery cohort: 112 (62 [12]); replicative cohort: 58 (59 [11])	..	Structural connectivity	Tract density index	The number of fibres encompassing the lesion is associated with survival	+++
Wei et al (2023) ³¹	Glioblastoma	Discovery cohort: 117 (59); validation cohort: 42 (59)	117 (60)	Structural connectivity	White matter tracts	Widespread structural impairment beyond the focal lesion is associated with survival	+++
Daniel et al (2021) ³²	Low-grade glioma and high-grade glioma	56 (58 [15])	30 (66 [11])	Functional connectivity	Homotopic connectivity	Disrupted connectivity in high-grade glioma is associated with survival	+
Daniel et al (2021) ³³	Glioblastoma	57 (58 [14])	100	Functional connectivity	Intranetwork connectivity	Higher connectivity is associated with better survival	+
Lamichhane et al (2021) ³⁴	Glioblastoma	64	..	Functional connectivity	Functional connectivity correlations	Functional connectivity features used in a support vector machine model can predict survival classes	++
Luckett et al (2023) ³⁵	Glioblastoma	133 (61 [12])	..	Functional connectivity	Functional connectivity correlations	A deep neural network trained with functional connectivity (together with other clinical factors) was accurate in classifying survival (<1 year, 1–2 years, >2 years)	++
Park et al (2023) ³⁶	Glioblastoma	189 (61 [11])	189 (68 [11])	Functional connectivity	Frequency analysis	Changes in spectra content of spontaneous fMRI fluctuations relate to different survival classes	+
Sprugnoli et al (2022) ³⁷	Low-grade glioma and high-grade glioma	54 (51 [15])	..	Functional connectivity	Seed-tumour connectivity	Tumour-brain connectivity patterns positively correlate with survival	+
Stoecklein et al (2020) ³⁸	Low-grade glioma and high-grade glioma	34 (49 [17])	..	Functional connectivity	Connectivity abnormality index	Diffusion alterations in both perilesional and distal regions are associated with cognition, and marginally with survival	-
Mandal et al (2020) ³⁹	Low-grade glioma and high-grade glioma	355	..	Network topology	Indirect functional connectivity	Gliomas are more frequent in brain regions acting as functional connector hubs than in regions that do not act as functional connector hubs	..
Mandal et al (2021) ⁴⁰	Low-grade glioma and high-grade glioma	Independent cohort analysis: 242 (53 [15]); replicability sample: 168	..	Network topology	Lesion covariance network	Spatial correspondence between gliomas localisation and networks is not linked with survival	--
Romero-Garcia et al (2023) ⁴¹	Low-grade glioma and high-grade glioma	242 (53 [15])	..	Network topology	Indirect functional connectivity	Low-grade and high-grade gliomas occur in regions with different network topological properties	..
Sansone et al (2023) ⁴²	Glioblastoma	99 (62 [11])	..	Network topology	Lesion network overlap	Network-spatially distribution pattern of gliomas	--

+++ Indicates a significant association with overall survival with independent validation. ++ Indicates a significant association with in-sample validation. + Indicates a significant association without validation in an independent cohort. - Indicates a trend toward association. -- Indicates no significant associations. Survival class= differentiation between short-term and long-term survival.

Table: The relationship between brain connectome and overall survival, topography, and histological grade of a glioblastoma

Panel 2: The brain as a complex network

The brain is organised as a hierarchical structure, composed of pathways with functional interactions between brain regions. Structural and functional connectivity can be assessed in vivo at the macroscale level (mm to cm), with structural connectivity measured by use of diffuse-weighted imaging, and functional connectivity measured by use of fMRI. Structurally, the brain is characterised by a small world scale-free architecture, where many short-range connections coexist with few long-range connections. Functionally, brain regions interact closely following the structural architecture under conditions in which cognition functions are minimised, such as during anaesthesia or slow-wave sleep.⁵⁸ However, in the alert state and during cognitive activities, the spatial similarity between structure and function greatly weakens, as functional patterns become more variable and less constrained by anatomy. Functional patterns at rest replay patterns observed during active behaviour and are organised in hierarchical functional systems that span gradients, from sensory motor functions to cognitive functions.⁵⁹

clinical MRI scans and was more robustly associated with overall survival than other commonly used variables (performance status, surgical resection, or genetics), irrespective of the overall length of survival. Finally, this index could differentiate, with an accuracy of 87%, patients with shorter (<20 months) versus longer (>20 months) survival.¹⁶ Brain locations of high tract density index corresponded to long-range white matter tracts (eg, arcuate, superior longitudinal, callosum), which is consistent with the hypothesis of glioblastoma spread along white matter tracts. Collectively, these two studies offer complementary methods for predicting overall survival based on the effect of glioblastoma on the structural connectome. The first approach provides anatomical details regarding the affected tracts, whereas the tract density index condenses this pathogenetic link between structural connections and tumour spreading into a single number (density of white matter fibres) and

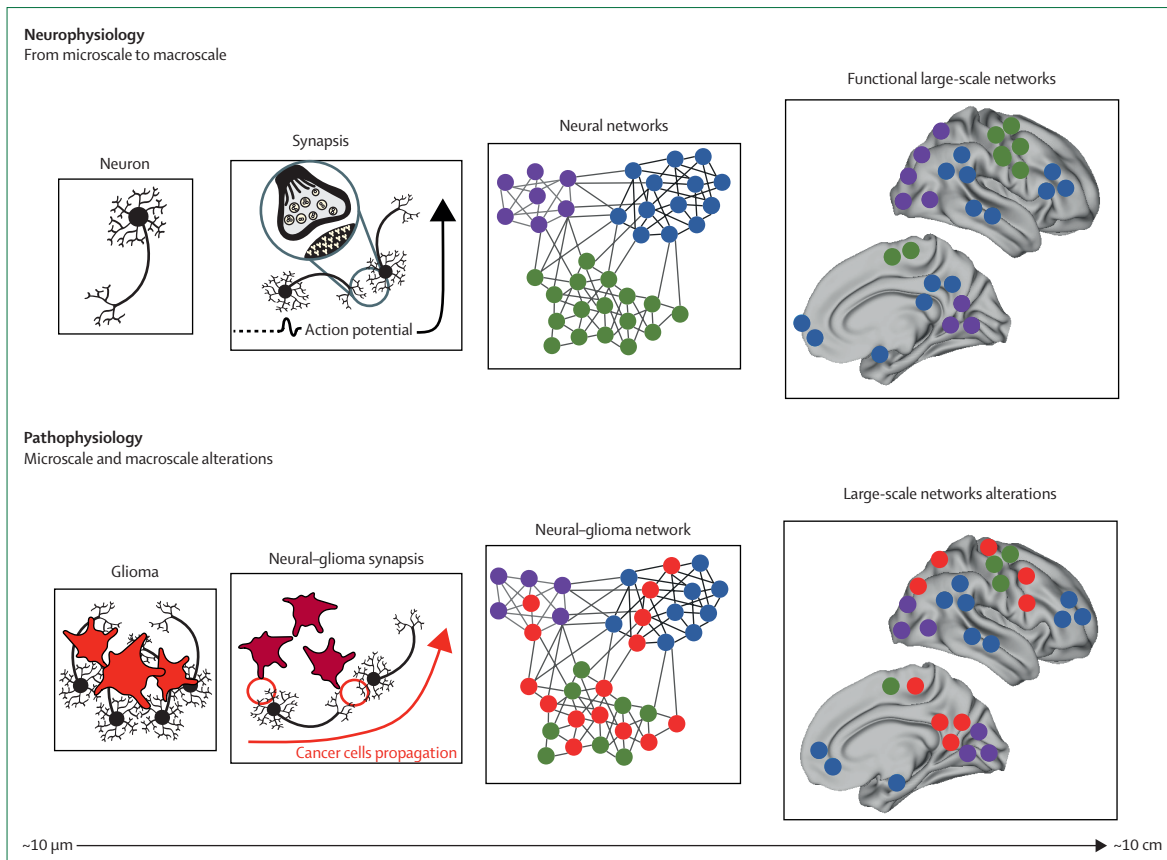


Figure 1: From the microscale to the macroscale investigation of glioma pathophysiology

Synaptic connections are established between neurons that, in turn, are organised into neural networks, and ultimately result in large-scale functional brain networks. Glioblastoma can alter this organisation by embedding into neural circuits and triggering alterations in brain networks.

it is obtainable from clinical MRI scans, thereby simplifying its clinical application. In both cases, the use of disconnection-derived measures resulted in increased accuracy in predicting overall survival (table).

Functional connectivity and carcinogenesis

Most resting-state fMRI studies of patients with brain tumours focus on a single network at a time (eg, language, default mode, or frontoparietal network), and have shown differences with healthy controls and correlations with the impairment of the corresponding functional domain (eg, language or attention).^{43,44} To date, only a few studies have applied a whole brain multinetwork approach to studying brain tumours, which has shown that functional networks far away from a glioblastoma lesion can be functionally altered (including regions in the contralateral hemisphere^{36,45–47}) and that multinetwork alterations can predict functional deficits.⁴⁵ The application of fMRI to the study of brain tumours has mainly focused on presurgical mapping and its role as a biomarker or predictor of cognitive impairment. However, evidence from the past 5 years highlights functional interconnections between the brain and glioblastoma at the synaptic level.^{48,49}

Neuronal activity-dependent paracrine signalling, through neuroligin-3 and BDNF, promotes glioma progression; furthermore, membrane depolarisation mediated by glutamatergic α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) receptors in neuron-to-glioma excitatory electrochemical synapses drives glioblastoma proliferation.^{49–53} Moreover, malignant cells can electrically integrate into neural circuits through neuron-to-glioma synapses and glioblastoma infiltrated brain tissue has functional connectivity (measured through short-range electrocorticography) with surrounding healthy tissue.^{50,54,55} Some brain regions affected by the glioblastoma have relatively higher levels of functional connectivity with regions distal from the tumour itself than other glioblastoma-affected regions. In these regions with increased connectivity, there is an upregulation of genes that are involved in the assembly of neural connections (NTNG1) and synaptogenesis (including thrombospondin-1).⁵⁵ The functional integration between neurons and the glioblastoma suggests new pharmacological strategies. For example, inhibition of thrombospondin-1 has been shown to decrease glioblastoma cell proliferation.⁵⁵ A phase 1/2 clinical trial of molecules that target the gap junction

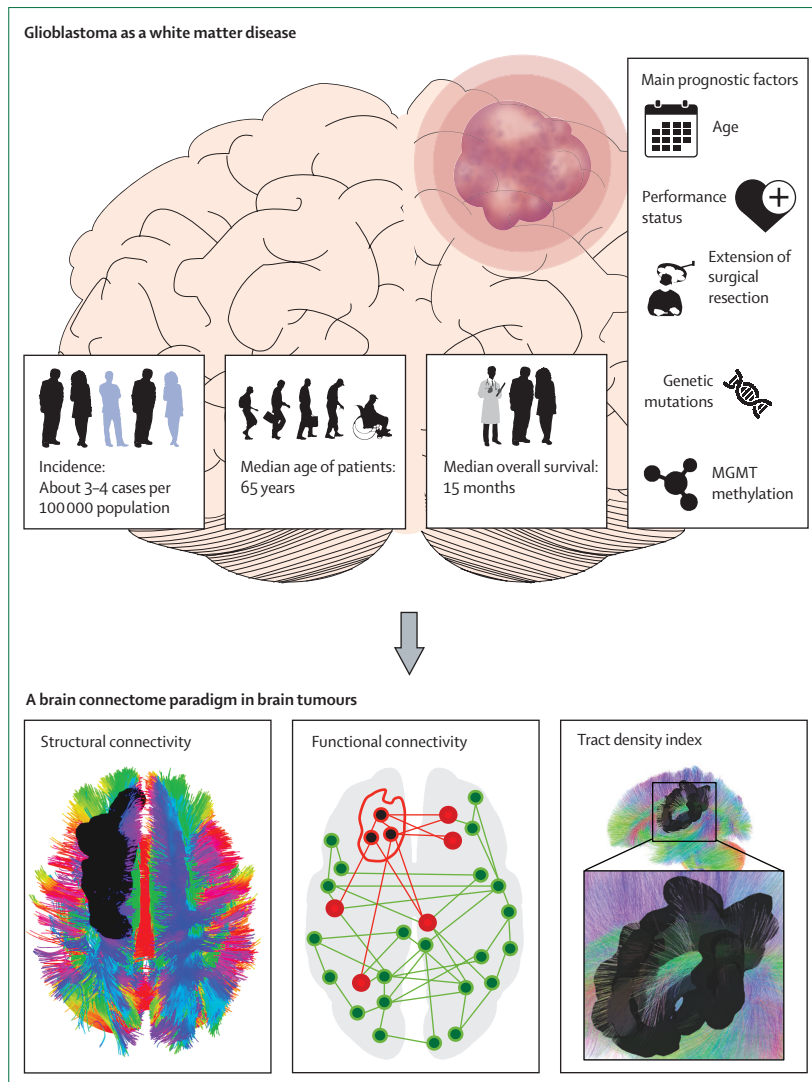


Figure 2: A brain connectome paradigm shift in glioblastoma

Traditional thinking on glioblastoma is based on the tumor itself without considering the active role of brain structural and functional intrinsic organisation (the connectome). Connectivity measures might provide new tools for assessing glioblastoma, such as the description of spreading pathways and a prognostic index (the tract density index).

(2021-000708-39) and neuron-to-glioma synapsis (2023-503938-52-00; AMPA receptors in the synapsis) are already registered.^{56,57}

This functional interaction between the CNS and the glioblastoma has an effect at the network level, with the formation of extensively interconnected neural-glioma networks, which possibly result in altered large-scale connectivity. Functional connectivity might then be a sensitive measure to link microscale (glioma-neural coupling) with macroscale (large-scale network) alterations induced by glioblastoma proliferation (panel 2, figure 1). As neuronal activity is a promoter of cancer cell proliferation and growth, functional connectivity measures could be a proxy of this mechanism and act as a prognostic predictor.

Unlike the focal injury of patients with stroke—in whom the fMRI signal within the lesion is disrupted by the ischaemic damage that destroys neurons, glia, and blood vessels and leaves what is essentially a hole in the brain—the brain tissue within and near a glioblastoma lesion is displaced but structurally intact. The biological significance of fMRI signals near or at a glioblastoma lesion could be the result of multiple factors. Glioblastomas can displace astrocytic processes and disrupt the neurovascular unit, thus interfering with fMRI signals.^{60,61} Conversely, glioblastoma might enhance synaptic activity through a bidirectional neuron-to-glioma interconnection, resulting in an increase (if excitatory pathways) or decrease (if inhibitory pathways) in functional connectivity. Studies measuring functional connectivity within the tumour have shown that a reduction of functional connectivity between the glioblastoma site and the homologous contralateral regions (homotopic connectivity) was associated with shorter overall survival,³² whereas higher functional connectivity was associated with longer overall survival.³³ Additionally, functional connectivity with distant brain regions showed significant correlation with overall survival, with the nature of the correlation being positive or negative depending on the distant brain region.^{36,37} These findings could be interpreted by considering that, if a glioblastoma affects the structural connectome at a distance, then it should also cause a corresponding decrease of functional communication with those regions. This interpretation is based on the relationship between structural and functional connectivity observed both in healthy subjects,⁶² and in patients with focal stroke lesions.⁶³ Relative increases in functional connectivity can reflect a secondary adjustment of connectivity in distant brain regions (as has been also shown in patients with stroke)⁶⁴ or, less likely, enhanced connectivity with the tumour site. There is clear evidence of distant intrahemispheric and interhemispheric alterations to both structural and functional connectivity in patients with glioblastoma.^{31,45} However, it is not known how these alterations relate to each other. Functional connectivity changes that reflect abnormalities in synchrony might precede structural connectivity alterations.

Neuronal activity plays a role not only in the progression and prognosis of gliomas, but also in their initiation. In a mouse model of optic pathway glioma, the stimulation of optic nerve activity increased tumour growth, whereas light deprivation prevented tumour formation.⁴⁸ Glioblastoma preferentially locates in correspondence of certain functional networks, with a higher prevalence in associative networks.^{39,41} More precisely, glioblastomas show a preference for locating in correspondence of connector hubs—brain regions that are functionally connected with multiple networks and are important for the integration of cognitive functions.^{65,66} There is a significant association between the location of the glioblastoma, the location of connectivity hubs, and the location of gene expression related to metabolic activity,

synaptic signalling, and gliomagenesis.^{39–42} In the brain regions where glioblastomas more commonly develop, the most highly expressed genes are related to cellular functions such as chromatin organisation, endosomal transport, G0 to G1 transition, metabolic processes, and synaptic transmission.⁴¹ Taken together, these observations suggest that brain regions containing many synapses will have higher metabolic demands, and a higher turnover of glial cells, which will in turn influence the risk of brain cancer.

Although we emphasise the importance of topology in overall survival, it is important to remember that overall survival also relates to the heterogeneity of glioblastoma in terms of molecular and cellular characteristics, and of the tumour itself and the microenvironment. At the macroscopic level, this heterogeneity could manifest in different percentages of necrosis, contrast enhancing or non-contrast enhancing, and oedema tissue, which might in turn reflect different microscopic features. For instance, in patients with glioblastoma, MRI radiomic features are able to identify regional biological signatures, including immune infiltration.^{67,68}

Potential therapeutic approaches

Surgical approaches try to strike a balance between complete tumour removal and preservation of neurological function, often referred to as maximal safe resection. It has been proposed that, especially in patients with specific molecular profiles, the resection borders should extend beyond the edges of contrast-enhanced tissue.⁶⁹ Connectivity measures could offer surgeons reliable prognostic indexes before surgery. Nowadays, connectivity measures such as diffusion tensor imaging are used to spare eloquent regions. By knowing the most common fibre tracts affected by a glioblastoma lesion, or the local areas near or at the glioblastoma with higher TDI, a more secure and in-depth resection might be feasible than is currently done. White matter or functional connectivity indices could then be uploaded to neurosurgical navigation systems, along with intraoperative imaging and neurophysiological monitoring, and could enhance the extent of the resection and reduce perioperative morbidity when aiming for a supratotal resection.⁷⁰

Radiotherapy, in conjunction with chemotherapy, stands as the gold standard for the treatment of glioblastoma. The target region for radiation includes a 20 mm border around the T1 contrast enhancement.⁷¹ However, the brain tissue is not uniform around this border; it varies in terms of white matter tract density, direction, and type. Consequently, the risk of malignant spreading is not the same across the border. As a result, the radiation target might be suboptimal and could be substantially enhanced by adding information on the structural connectome.

Viewing glioblastoma not as an isolated entity but as an entity embedded within the connectome opens the door to expanding treatment options through approaches based on brain network connectivity (figure 2). For instance,

Panel 3: Glioblastoma as a white matter disease: a paradigm shift

Centuries ago, Galen postulated that the transmission of animal spirits through interconnected pathways could facilitate the spread of brain diseases. Fast forward almost two millennia, and we can note a parallel mechanism in glioblastoma if replacing the antiquated concept of animal spirits with that of cancer cells. The burgeoning body of evidence supporting the structural and functional interplay between the brain and glioma has given rise to the field of cancer neuroscience, heralding a paradigm shift.

Perseverating with an exclusive oncological viewpoint on glioblastoma is now outdated. Strategies effective for other cancers prove inadequate for brain cancer, evident in the dearth of noteworthy improvements in patients' outcomes.^{5,6} Brain cancer is often treated as a generic form of cancer, focusing primarily on impeding the proliferation of cancer cells and suppressing neoangiogenesis. However, this approach tends to overlook the crucial role played by the intricate connectivity of the brain in fostering tumour growth and dissemination. To reframe our understanding, glioblastoma should be recognised as a neurological disease, specifically a white matter disease, originating from astrocytes and predominantly thriving in the frontal, temporal, and parietal white matter regions.³⁹ Additionally, the primary dissemination route of cancer cells aligns with structural anatomical pathways.^{25–27}

This paradigm shift necessitates a fresh perspective and demands increased involvement from neuroscientists. Evidence of brain-glioma interaction at the synaptic or local network level has identified new potential pharmacological targets,^{56,57} but a brain connectivity approach could provide new prognostic indexes and suggest further therapeutic strategies. Existing evidence underscores the pivotal role of connectivity-derived measures in prognostication, which outperform conventional prognostic factors.^{16,31,37} Advancements in the assessment of the blood oxygen level-dependent signal within white matter might lead to substantial contributions in this field.⁷⁵

non-invasive brain stimulation is used to safely modulate brain activity and plasticity using electromagnetic or electrical stimulation. Non-invasive brain stimulation can be used to modulate or inhibit synaptic signalling in neurons, potentially interfering with pathological neuron-to-glioma synaptogenesis. This interface can lead to the suppression of synaptic signalling in neurons surrounding a glioblastoma, ultimately slowing down the mitosis and migration rate of tumoural cells.^{53,72,73} Additionally, non-invasive brain stimulation has non-neuronal effects, including the modulation of perfusion and blood–brain barrier permeability, potentially enhancing drug delivery. Although these approaches are still speculative, if tumour treating fields improve progression-free survival despite

Search strategy and selection criteria

References for this Personal View were identified by searches of PubMed for articles published from January, 2020, to January, 2024, with multiple combinations of search terms: "glioma", "glioblastoma", "structural connectivity", "functional connectivity", "connectome", and "survival". We selected articles if they reported on the relationship between glioblastoma and brain connectivity. We also considered publications cited in these articles that did not appear in the search algorithm, and articles published at an earlier date (if deemed relevant). The final selection of references reflects our subjective assessment of their relevance to the topics covered in this Personal View and with respect to the results reported in these publications and their methodological quality.

the lack of anatomical specificity,⁷⁴ strategies aiming to selectively stimulate only tumour-related networks might achieve even better outcomes.

Conclusions and future directions

A dynamic, bidirectional interplay unfolds between cancer cells and neurons, challenging the notion that glioblastoma merely resides passively in the brain. Instead, the tumour actively engages with both the structural and functional organisation of the brain. Delving into the intricacies of this structural and functional connectivity will unveil a deeper understanding of glioblastoma and enhance prognosis. The reliability of connectivity-derived measures as prognostic factors has been robustly shown, prompting their integration into clinical practice. Furthermore, their inclusion in clinical trials might prove valuable for the stratification of participants. Taking the next logical step, connectivity-derived measures could be integrated into surgical and radiation treatments, potentially revolutionising therapeutic approaches.

Large open access datasets of glioblastoma (including brain connectivity data) will hasten progress in this field. Similarly, this paradigm shift would foster collaborations between neurologists studying various neurological disorders, such as proteinopathies or stroke, in which brain connectivity alterations also play a major role. An important initiative at the European level is the development of a digital infrastructure called EBRAINS, where large datasets will be shared and analysed to extract imaging features associated with cognition, survival, and disability. Surprisingly, however, none of the current international guidelines on brain tumours incorporates the study of brain connectivity. Whether this status quo evolves in the coming years hinges on the concerted efforts of clinicians and researchers to bridge the gap between oncology and neuroscience. This endeavour necessitates a paradigm shift (panel 3), transcending the exclusive focus on cancer and embracing a holistic perspective that considers the intricate interaction between the tumour and its host: the brain.

Contributors

Conceptualisation: AS, MC; literature search and writing: AS, LP; figure design: LP; review and editing: AB, MC; funding acquisition for research facilities and personnel: MC.

Declaration of interests

AS, LP, and MC report a patent pending (102022000015360; PCT IB2023/057357) for the method to compute the tract density index ("Un nuovo indice prognostico di sopravvivenza nel glioblastoma multiforme basato sulle alterazioni delle connessioni della sostanza bianca cerebrale misurato con tecniche di diffusione RM"). AB declares no competing interests.

Acknowledgments

MC was supported by Fondazione Cassa di Risparmio di Padova e Rovigo (CARIPARO)—Ricerca Scientifica di Eccellenza 2018 (Grant Agreement number 55403); Italian Ministero della Salute, Brain connectivity measured with high-density electroencephalography: a novel neurodiagnostic tool for stroke (NEUROCONN; RF-2018-1236689); Horizon 2020 European School of Network Neuroscience - European School of Network Neuroscience (euSNN), H2020-SC5-2019-2 (Grant Agreement number 860563); Horizon 2020 research and innovation programme; Visionary Nature Based Actions For Health, Wellbeing & Resilience in Cities (VARCITIES), Horizon 2020-SC5-2019-2 (Grant Agreement number 869505); Italian Ministero della Salute: Eye-movement dynamics during free viewing as biomarker for assessment of visuospatial functions and for closed-loop rehabilitation in stroke (EYEMOVINSTROKE; RF-2019-12369300); HORIZON-ERC-SyG (Grant No.101071900).

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